

Лабораторная работа №8

Адресация IPv4 и IPv6. Настройка маршрутизации

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2025

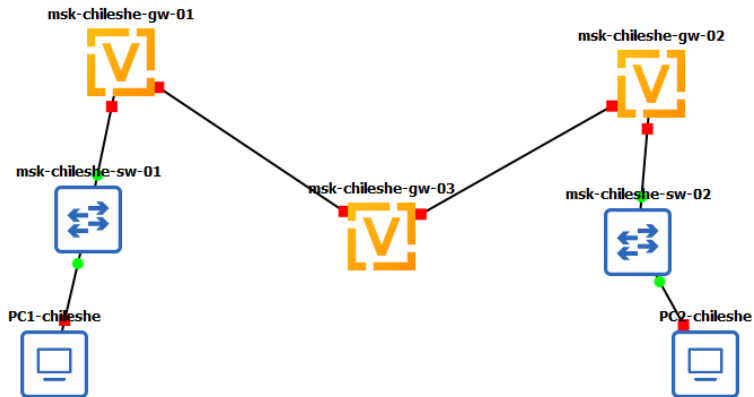
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Цели и задачи работы

Изучение принципов маршрутизации в IPv4- и IPv6-сетях
и практическая настройка сетевого оборудования в среде GNS3.

- Смоделировать сеть на базе VyOS и коммутаторов в GNS3
- Настроить IPv6-адресацию конечных узлов и RA на маршрутизаторах
- Проверить связность до и после настройки динамической IPv4-маршрутизации (RIP)
- Создать туннель IPv6 over IPv4 (SIT) и настроить статическую маршрутизацию IPv6
- Подтвердить работу сети проверками ping/trace и анализом трафика в Wireshark

Выполнение работы



- LAN1 (IPv6): 1000::/64 (PC1 — 1000::a/64, GW — 1000::1/64)
- LAN2 (IPv6): 1002::/64 (PC2 — 1002::a/64, GW — 1002::1/64)
- Транзит IPv4: 10.0.0.0/8 (между gw-01 и gw-03)
- Транзит IPv4: 20.0.0.0/8 (между gw-03 и gw-02)

Настройка PC1 (IPv6)

```
PC1-chileshe - PuTTY
Executing the startup file

PC1-chileshe> ip 1000::a/64
PC1 : 1000::a/64

PC1-chileshe> save
Saving startup configuration to startup.vpc
. done

PC1-chileshe> show ip

NAME           : PC1-chileshe[1]
IP/MASK        : 0.0.0.0/0
GATEWAY        : 0.0.0.0
DNS            :
MAC            : 00:50:79:66:68:00
LPORT          : 10012
RHOST:PORT     : 127.0.0.1:10013
MTU            : 1500

PC1-chileshe> show ipv6

NAME           : PC1-chileshe[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE     : 1000::a/64
DNS             :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT          : 10012
RHOST:PORT     : 127.0.0.1:10013
MTU            : 1500

PC1-chileshe>
```


Настройка PC2 (IPv6)

```
PC2-chileshe - PuTTY

Executing the startup file

PC2-chileshe> ip 1002::a/64
PC1 : 1002::a/64

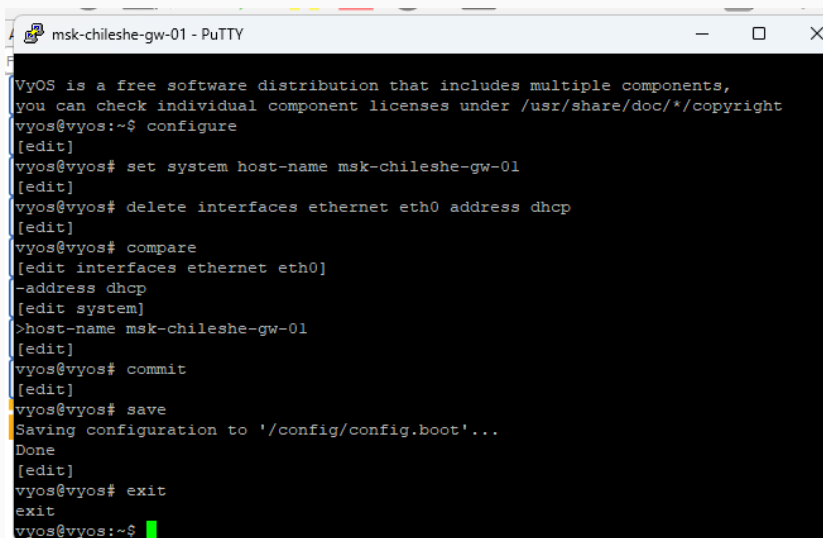
PC2-chileshe> save
Saving startup configuration to startup.vpc
. done

PC2-chileshe> show ipv6

NAME                : PC2-chileshe[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10014
RHOST:PORT          : 127.0.0.1:10015
MTU                 : 1500

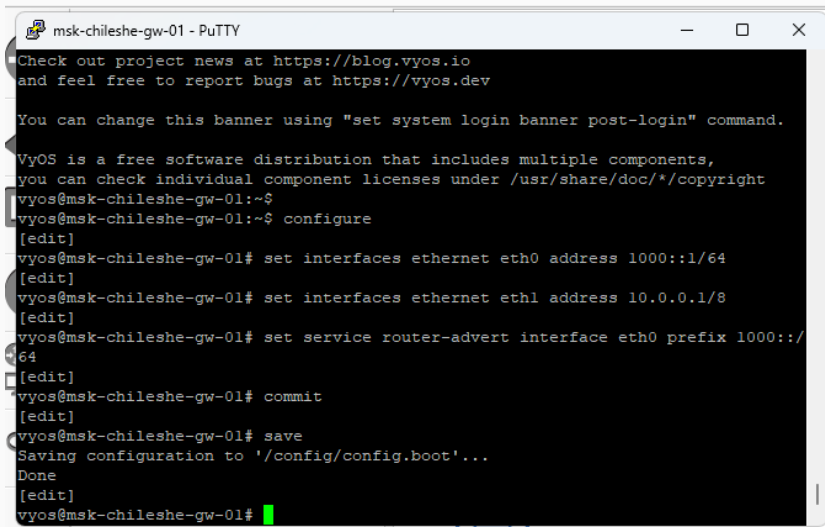
PC2-chileshe> 
```

Идентификация маршрутизатора



```
msk-chileshe-gw-01 - PuTTY
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-chileshe-gw-01
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
-address dhcp
[edit system]
>host-name msk-chileshe-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$
```

Рис. 4: Изменение имени маршрутизатора msk-chileshe-gw-01

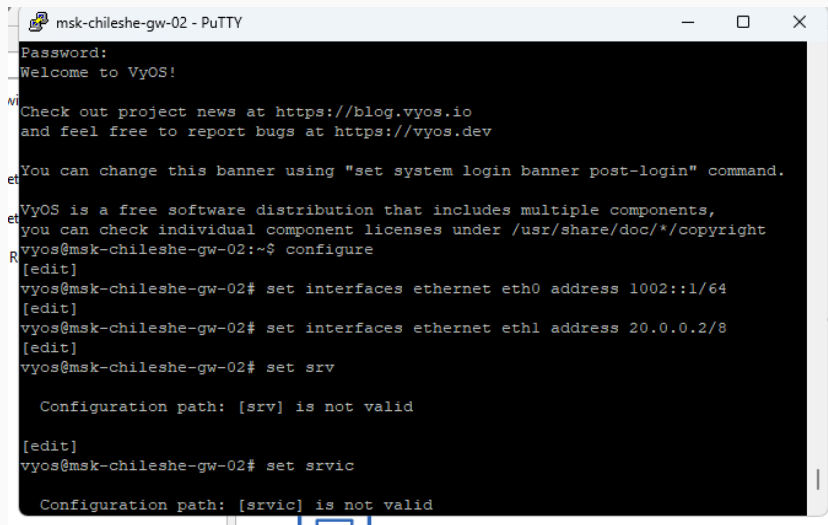


```
msk-chileshe-gw-01 - PuTTY
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-chileshe-gw-01:~$
vyos@msk-chileshe-gw-01:~$ configure
[edit]
vyos@msk-chileshe-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-chileshe-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
vyos@msk-chileshe-gw-01# set service router-advert interface eth0 prefix 1000::/
64
[edit]
vyos@msk-chileshe-gw-01# commit
[edit]
vyos@msk-chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-01#
```

Рис. 5: Настройка интерфейсов msk-chileshe-gw-01



```
msk-chileshe-gw-02 - PuTTY
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-chileshe-gw-02:~$ configure
[edit]
vyos@msk-chileshe-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-chileshe-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8
[edit]
vyos@msk-chileshe-gw-02# set srv

Configuration path: [srv] is not valid

[edit]
vyos@msk-chileshe-gw-02# set srvic

Configuration path: [srvic] is not valid
```

Рис. 6: Настройка интерфейсов msk-chileshe-gw-02

```
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-chileshe-gw-03:~$ configure
[edit]
vyos@msk-chileshe-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8
[edit]
vyos@msk-chileshe-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
vyos@msk-chileshe-gw-03# commit
[edit]
vyos@msk-chileshe-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-03#
```

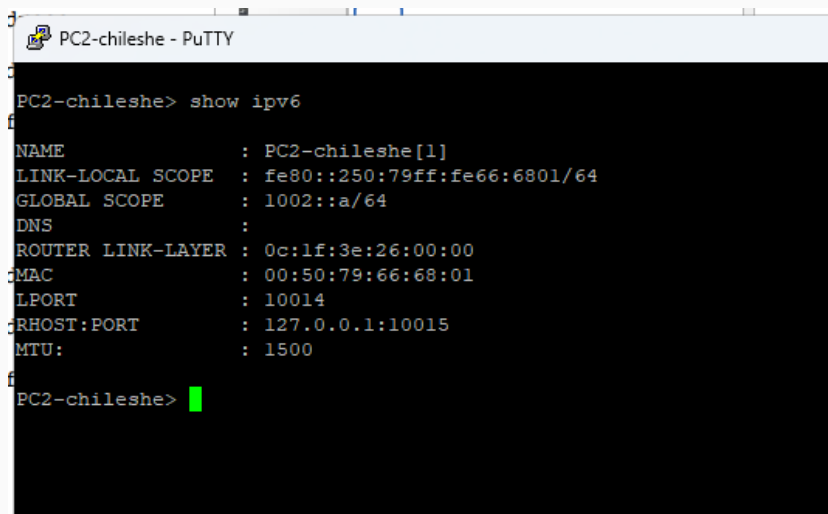
Рис. 7: Настройка интерфейсов msk-chileshe-gw-03

- eth0: 10.0.0.2/8
- eth1: 20.0.0.1/8

```
PC1-chileshe>  
PC1-chileshe> show ipv6  
  
NAME                : PC1-chileshe[1]  
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64  
GLOBAL SCOPE        : 1000::a/64  
DNS                 :  
ROUTER LINK-LAYER   : 0c:d3:a2:84:00:00  
MAC                 : 00:50:79:66:68:00  
LPORT              : 10012  
RHOST:PORT          : 127.0.0.1:10013  
MTU                 : 1500  
  
PC1-chileshe>
```

Рис. 8: Проверка IPv6 на PC1

- Появился ROUTER LINK-LAYER (MAC ближайшего маршрутизатора)
- Это подтверждает работу RA и NDP в сегменте 1000::/64



```
PC2-chileshe> show ipv6

NAME                : PC2-chileshe[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                  :
ROUTER LINK-LAYER   : 0c:1f:3e:26:00:00
MAC                  : 00:50:79:66:68:01
LPORT                : 10014
RHOST:PORT           : 127.0.0.1:10015
MTU                  : 1500

PC2-chileshe> 
```

Рис. 9: Проверка IPv6 на PC2

Проверка связности с gw-01

```
exit
vyos@msk-chileshe-gw-01:~$
vyos@msk-chileshe-gw-01:~$ ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.61 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.564 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=1.53 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6ms
rtt min/avg/max/mdev = 0.564/1.567/2.608/0.836 ms
vyos@msk-chileshe-gw-01:~$ ping 20.0.0.1
connect: Network is unreachable
vyos@msk-chileshe-gw-01:~$ ping 20.0.0.2
connect: Network is unreachable
vyos@msk-chileshe-gw-01:~$
```

Рис. 10: Проверка связности до настройки RIP

Наблюдения: - 10.0.0.2 доступен (непосредственно подключённая сеть 10.0.0.0/8)

- 20.0.0.1 и 20.0.0.2 недоступны (Network is unreachable)

Причина: отсутствует маршрут к 20.0.0.0/8

Анализ трафика до RIP

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:d3:a2:84:00:01	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
2	0.000916	0c:d3:92:48:00:00	0c:d3:a2:84:00:01	ARP	60	10.0.0.2 is at 0c:d3:92:48:00:00
3	0.001346	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0852, seq=1/256, ttl=64 (reply 3
4	0.001870	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0852, seq=1/256, ttl=64 (request
5	1.001869	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0852, seq=2/512, ttl=64 (reply 3
6	1.002180	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0852, seq=2/512, ttl=64 (request
7	2.003360	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0852, seq=3/768, ttl=64 (reply 3
8	2.004030	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0852, seq=3/768, ttl=64 (request
9	5.494136	0c:d3:92:48:00:00	0c:d3:a2:84:00:01	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2
10	5.495145	0c:d3:a2:84:00:01	0c:d3:92:48:00:00	ARP	60	10.0.0.1 is at 0c:d3:a2:84:00:01

Frame 8: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0	
Ethernet II, Src: 0c:d3:92:48:00:00 (0c:d3:92:48:00:00), Dst: 0c:d3:a2:84:00:01 (0c:d3:a2:84:00:01)	
Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.2	
Internet Control Message Protocol	
Type: Echo (ping) reply (0)	0000 0c d3 a2 84 00 01 0c d3 92 48 00 00 08 00 45 00
Code: 0	0010 00 54 30 5a 00 00 40 01 36 4d 0a 00 00 02 0a 00
Checksum: 0x7318 [correct]	0020 00 01 00 00 73 18 08 52 00 03 74 bf 9e 69 00 00
[Checksum Status: Good]	0030 00 00 b0 96 02 00 00 00 00 00 10 11 12 13 14 15
Identifier (BE): 2130 (0x0852)	0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25
Identifier (LE): 21000 (0x5208)	0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35
Sequence Number (BE): 3 (0x0003)	0060 36 37
Sequence Number (LE): 768 (0x0300)	
[Request frame: 7]	
[Response time: 0,670 ms]	
ICMP Data: 74bf9e6900000000b096020000000010111213141516171819	

Рис. 11: Анализ ICMP-трафика до настройки RIP

```
[edit]
vyos@msk-chileshe-gw-01# set protocols rip netwo
; network          network-distance
[edit]
vyos@msk-chileshe-gw-01# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-chileshe-gw-01# commit
[edit]
vyos@msk-chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-01#
```

Рис. 12: Настройка RIP на msk-chileshe-gw-01

- Анонсируется сеть 10.0.0.0/8

```
[edit]  
vyos@msk-chileshe-gw-02#  
[edit]  
vyos@msk-chileshe-gw-02# set protocols rip network 20.0.0.0/8  
[edit]  
vyos@msk-chileshe-gw-02# commit  
[edit]  
vyos@msk-chileshe-gw-02# save  
Saving configuration to '/config/config.boot'...  
Done  
[edit]  
vyos@msk-chileshe-gw-02#
```

Рис. 13: Настройка RIP на msk-chileshe-gw-02

- Анонсируется сеть 20.0.0.0/8

```
[edit]
vyos@msk-chileshe-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-chileshe-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-chileshe-gw-03# commit
[edit]
vyos@msk-chileshe-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-03#
```

Рис. 14: Настройка RIP на msk-chileshe-gw-03

- Анонсируются сети 10.0.0.0/8 и 20.0.0.0/8
- gw-03 играет роль транзитного маршрутизатора между доменами

Повторная проверка связности с gw-01

```
msk-chileshe-gw-01 - PuTTY
[edit]
vyos@msk-chileshe-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.10 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=3.82 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=2.32 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6ms
rtt min/avg/max/mdev = 1.096/2.411/3.823/1.116 ms
[edit]
vyos@msk-chileshe-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=1.07 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=3.45 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=1.89 ms
^C
--- 20.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 5ms
rtt min/avg/max/mdev = 1.073/2.137/3.446/0.984 ms
[edit]
vyos@msk-chileshe-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=3.73 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=3.94 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=4.76 ms
^C
--- 20.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6ms
```

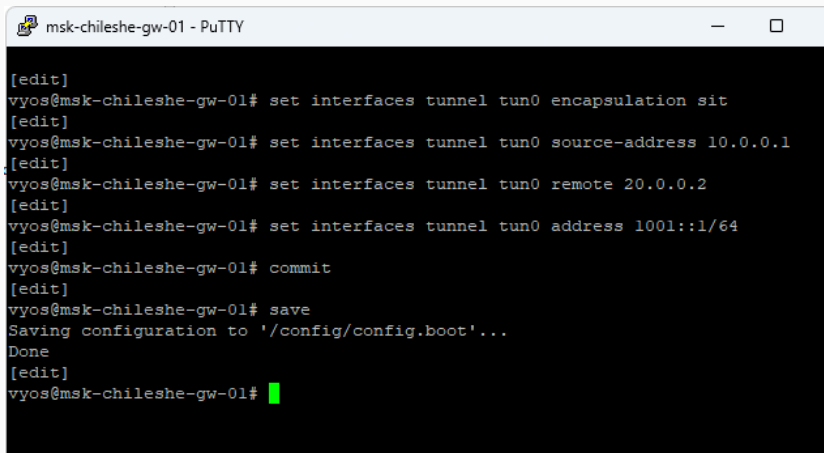
Трафик RIP и ICMP в Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.2	224.0.0.9	RIPv2	66	Response
2	2.908730	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x095f, seq=1/256, ttl=64 (reply 3
3	2.909396	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x095f, seq=1/256, ttl=64 (request 3
4	3.912965	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x095f, seq=2/512, ttl=64 (reply 3
5	3.914093	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x095f, seq=2/512, ttl=64 (request 3
6	4.912799	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x095f, seq=3/768, ttl=64 (reply 3
7	4.914270	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x095f, seq=3/768, ttl=64 (request 3
8	8.218207	0c:d3:a2:48:00:00	0c:d3:a2:84:00:01	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2
9	8.220922	0c:d3:a2:84:00:01	0c:d3:a2:48:00:00	ARP	60	10.0.0.1 is at 0c:d3:a2:84:00:01
10	8.271745	0c:d3:a2:84:00:01	0c:d3:a2:48:00:00	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
11	8.272991	0c:d3:a2:48:00:00	0c:d3:a2:84:00:01	ARP	60	10.0.0.2 is at 0c:d3:a2:84:00:01
12	12.254261	10.0.0.1	20.0.0.1	ICMP	98	Echo (ping) request id=0x0960, seq=1/256, ttl=64 (reply 3
13	12.254858	20.0.0.1	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0960, seq=1/256, ttl=64 (request 3
14	13.256270	10.0.0.1	20.0.0.1	ICMP	98	Echo (ping) request id=0x0960, seq=2/512, ttl=64 (reply 3
15	13.257682	20.0.0.1	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0960, seq=2/512, ttl=64 (request 3
16	14.257764	10.0.0.1	20.0.0.1	ICMP	98	Echo (ping) request id=0x0960, seq=3/768, ttl=64 (reply 3
17	14.258741	20.0.0.1	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0960, seq=3/768, ttl=64 (request 3
18	19.933539	10.0.0.1	20.0.0.2	ICMP	98	Echo (ping) request id=0x0961, seq=1/256, ttl=64 (reply 3
19	19.936647	20.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0961, seq=1/256, ttl=63 (request 3
20	22.801224	10.0.0.1	20.0.0.2	ICMP	98	Echo (ping) request id=0x0961, seq=2/512, ttl=64 (reply 3
21	22.804246	20.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0961, seq=2/512, ttl=63 (request 3
22	23.803454	10.0.0.1	20.0.0.2	ICMP	98	Echo (ping) request id=0x0961, seq=3/768, ttl=64 (reply 3
23	23.806786	20.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0961, seq=3/768, ttl=63 (request 3
24	37.869330	10.0.0.2	224.0.0.9	RIPv2	66	Response

Рис. 16: Анализ RIP и ICMP-трафика

- RIP передаётся на 224.0.0.9 (RIPv2)
- На фоне обновлений RIP видны ICMP-пакеты, проходящие через транзит

Создание туннеля на gw-01

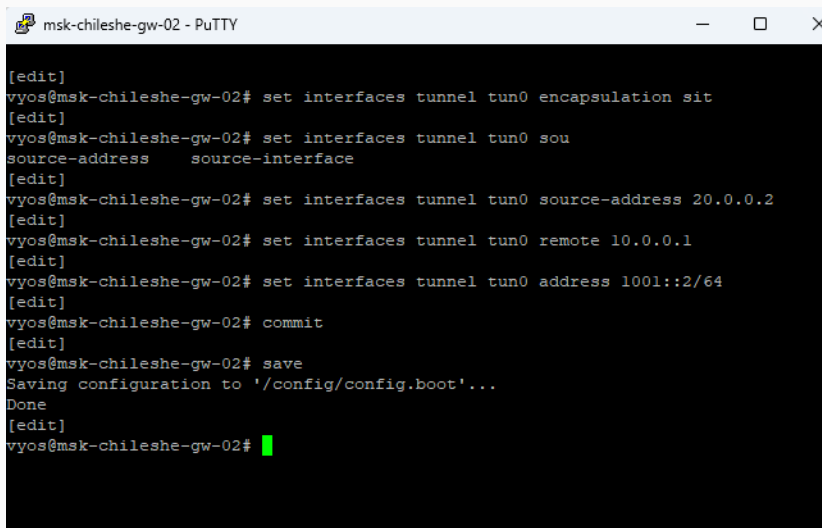


```
[edit]
vyos@msk-chileshe-gw-01# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-chileshe-gw-01# set interfaces tunnel tun0 source-address 10.0.0.1
[edit]
vyos@msk-chileshe-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
[edit]
vyos@msk-chileshe-gw-01# set interfaces tunnel tun0 address 1001::1/64
[edit]
vyos@msk-chileshe-gw-01# commit
[edit]
vyos@msk-chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-01#
```

Рис. 17: Настройка туннеля на msk-chileshe-gw-01

- tun0: SIT, source 10.0.0.1, remote 20.0.0.2

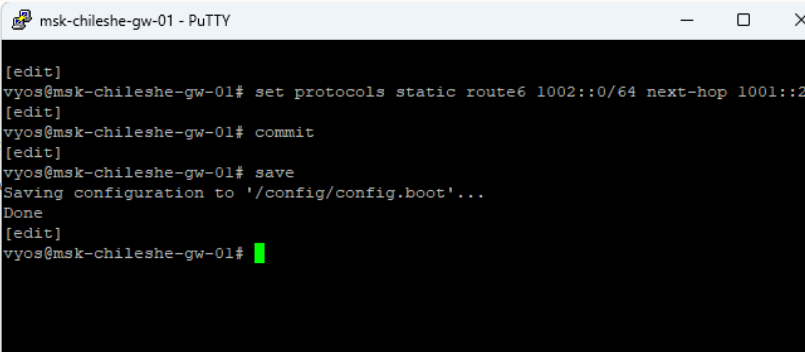
Создание туннеля на gw-02



```
[edit]
vyos@msk-chileshe-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-chileshe-gw-02# set interfaces tunnel tun0 sou
source-address      source-interface
[edit]
vyos@msk-chileshe-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-chileshe-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-chileshe-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-chileshe-gw-02# commit
[edit]
vyos@msk-chileshe-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-02#
```

Рис. 18: Настройка туннеля на msk-chileshe-gw-02

Route6 на gw-01

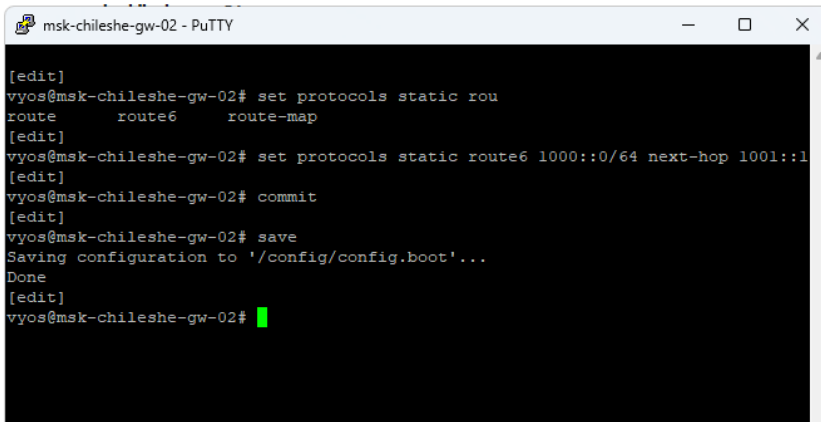


```
msk-chileshe-gw-01 - PuTTY
[edit]
vyos@msk-chileshe-gw-01# set protocols static route6 1002::0/64 next-hop 1001::2
[edit]
vyos@msk-chileshe-gw-01# commit
[edit]
vyos@msk-chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-01#
```

Рис. 19: Статический маршрут IPv6 на msk-chileshe-gw-01

- Маршрут к 1002::/64 через next-hop 1001::2 (туннель)

Route6 на gw-02



```
msk-chileshe-gw-02 - PuTTY
[edit]
vyos@msk-chileshe-gw-02# set protocols static rou
route      route6      route-map
[edit]
vyos@msk-chileshe-gw-02# set protocols static route6 1000::0/64 next-hop 1001::1
[edit]
vyos@msk-chileshe-gw-02# commit
[edit]
vyos@msk-chileshe-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-02#
```

Рис. 20: Статический маршрут IPv6 на msk-chileshe-gw-02

- Маршрут к 1000::/64 через next-hop 1001::1 (туннель)

PC1 → PC2 (ping и trace)

```
PC1-chileshe>
PC1-chileshe> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=6.433 ms
1002::a icmp6_seq=2 ttl=60 time=9.974 ms
1002::a icmp6_seq=3 ttl=60 time=9.444 ms
1002::a icmp6_seq=4 ttl=60 time=6.918 ms
1002::a icmp6_seq=5 ttl=60 time=7.014 ms

PC1-chileshe> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1    2.175 ms   1.592 ms   0.961 ms
 2 1001::2    3.659 ms   1.621 ms   1.708 ms
 3 1002::a    1.858 ms   1.378 ms   1.414 ms

PC1-chileshe>
```

Рис. 21: Ping и trace с PC1

- Успешные ответы ICMPv6
- Trace фиксирует прохождение через 1000::1 и туннельный next-hop 1001::2

```
PC2-chileshe> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=8.484 ms
1000::a icmp6_seq=2 ttl=60 time=9.635 ms
1000::a icmp6_seq=3 ttl=60 time=10.997 ms
1000::a icmp6_seq=4 ttl=60 time=6.583 ms
1000::a icmp6_seq=5 ttl=60 time=9.459 ms

PC2-chileshe> trace 1000::a

trace to 1000::a, 64 hops max
 1 1002::1    2.415 ms   1.445 ms   1.290 ms
 2 1001::1    4.079 ms   4.112 ms   2.330 ms
 3 1000::a    1.910 ms   1.986 ms   1.367 ms

PC2-chileshe> 
```

Инкапсуляция IPv6 в IPv4

Захват с Standard input [msk-chileshe-gw-01 eth1 to msk-chileshe-gw-03 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
24	29.889036	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xebc2, seq=1, hop limit=63 (reply in 25)
25	29.811894	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xebc2, seq=1, hop limit=61 (request in 24)
26	30.820073	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xebc2, seq=2, hop limit=63 (reply in 27)
27	30.823288	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xebc2, seq=2, hop limit=61 (request in 26)
28	31.833086	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xebc2, seq=3, hop limit=63 (reply in 29)
29	31.835856	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xebc2, seq=3, hop limit=61 (request in 28)
30	32.554121	10.0.0.2	224.0.0.9	RIPv2	66	Response
31	32.840887	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xebc2, seq=4, hop limit=63 (reply in 32)
32	32.842998	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xebc2, seq=4, hop limit=61 (request in 31)
33	33.849336	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xebc2, seq=5, hop limit=63 (reply in 34)
34	33.850819	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xebc2, seq=5, hop limit=61 (request in 33)

Frame 24: Packet, 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface -, id 0

Ethernet II, Src: 0c:3d:92:48:00:00 (0c:3d:92:48:00:00), Dst: 0c:d3:a2:84:00:01 (0c:d3:a2:84:00:01)

Internet Protocol Version 4, Src: 20.0.0.2, Dst: 10.0.0.1

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 124
- Identification: 0x3ab6 (15030)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 63
- Protocol: IPv6 (41)
- Header Checksum: 0xe2a0 [validation disabled]
[Header checksum status: Unverified]
- Source Address: 20.0.0.2
- Destination Address: 10.0.0.1
- [Stream index: 0]

Internet Protocol Version 6, Src: 1002::a, Dst: 1000::a

- 0110 = Version: 6
- 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
- 0000 0000 0000 0000 = Flow Label: 0x000000
- Payload Length: 64
- Next Header: ICMPv6 (58)
- Hop Limit: 63
- Source Address: 1002::a
- Destination Address: 1000::a
- [Stream index: 0]

Internet Control Message Protocol v6

- Type: Echo (ping) request (128)
- Code: 0

0000 0c d3 a2 84
0010 00 7c 3a b6
0020 00 01 60 00
0030 00 00 00 00
0040 00 00 00 00
0050 00 01 00 01
0060 0e 0f 10 11
0070 1e 1f 20 21
0080 2e 2f 30 31

Вывод

- Смоделирована сеть в GNS3 на базе VyOS и VPCS
- Настроены IPv6-адреса и Router Advertisement в локальных сегментах
- Динамическая IPv4-маршрутизация RIP обеспечила связность между 10.0.0.0/8 и 20.0.0.0/8
- Реализован туннель IPv6 over IPv4 (SIT) и статическая маршрутизация IPv6
- Доступность подтверждена ping/trace; инкапсуляция подтверждена анализом Wireshark